DSAA 4011 · LAB 9 · FRIDAY 6 NOVEMBER 2026

Weather on a Ring

Forecasting with learned messages: GraphCast as a toy

Global weather is now forecast by networks that pass messages on a graph. Today: a small version on a ring of stations, what it gets right, and where it fails.

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Builds on Lecture 9: learned message passing on graphs

The argument in five steps

- **The job: forecast 20 weather stations**

The trouble: a plain network breaks when we renumber the stations

The fix: one shared message rule on every link

The result: any size, any numbering, but it sees only as far as it passes messages

The frontier: long links, GraphCast, and attention

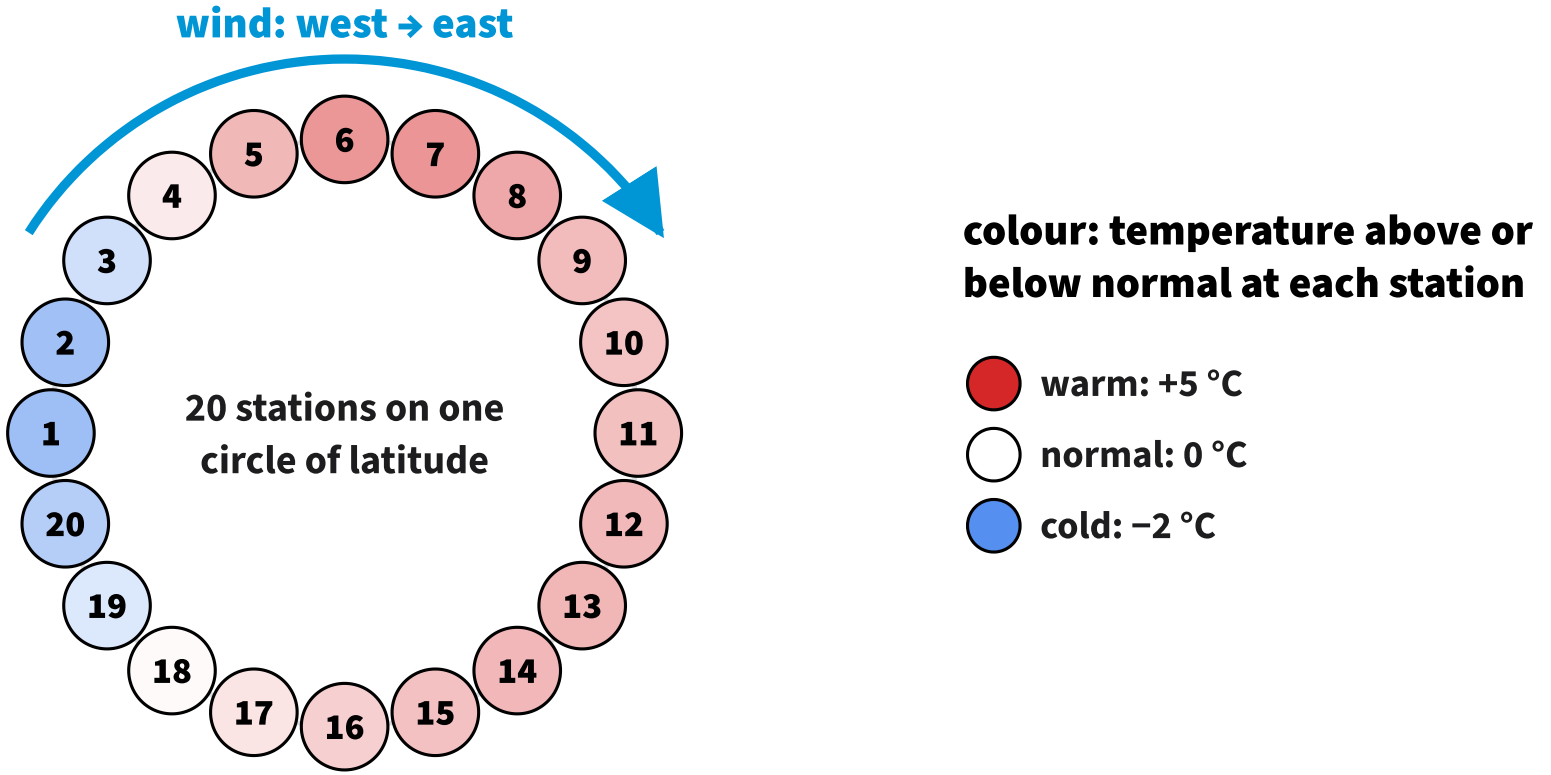
Each step answers the question the last one leaves open.

Main Question

A network learns to forecast from 20 weather stations. Can it forecast 100 stations, and how far ahead can we trust it?

1 · The job: forecast every station 6 hours ahead

Twenty stations sit on one circle of latitude; each reads its temperature.



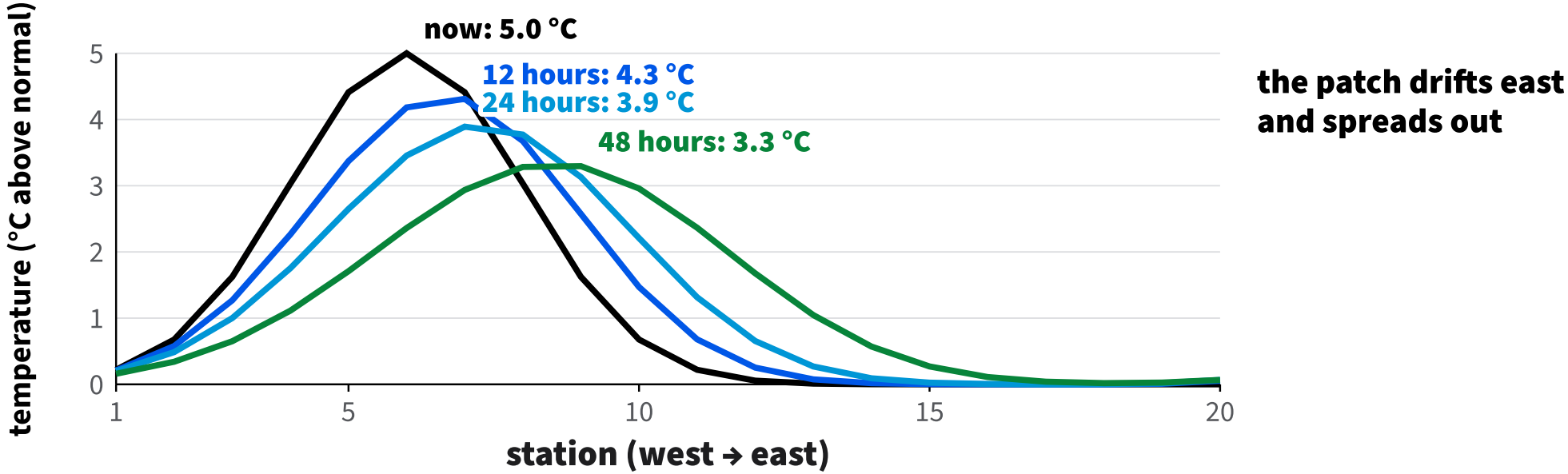
SO **From today's 20 readings, forecast all 20 readings 6 hours later.**

NEXT How does the toy weather change in 6 hours?

1 · The toy weather: a warm patch drifts east and spreads

Every 6 hours, each station mixes its temperature with its two neighbours'.

$$\text{new temperature} = 0.5 \cdot \text{west neighbour} + 0.3 \cdot \text{own} + 0.2 \cdot \text{east neighbour}$$



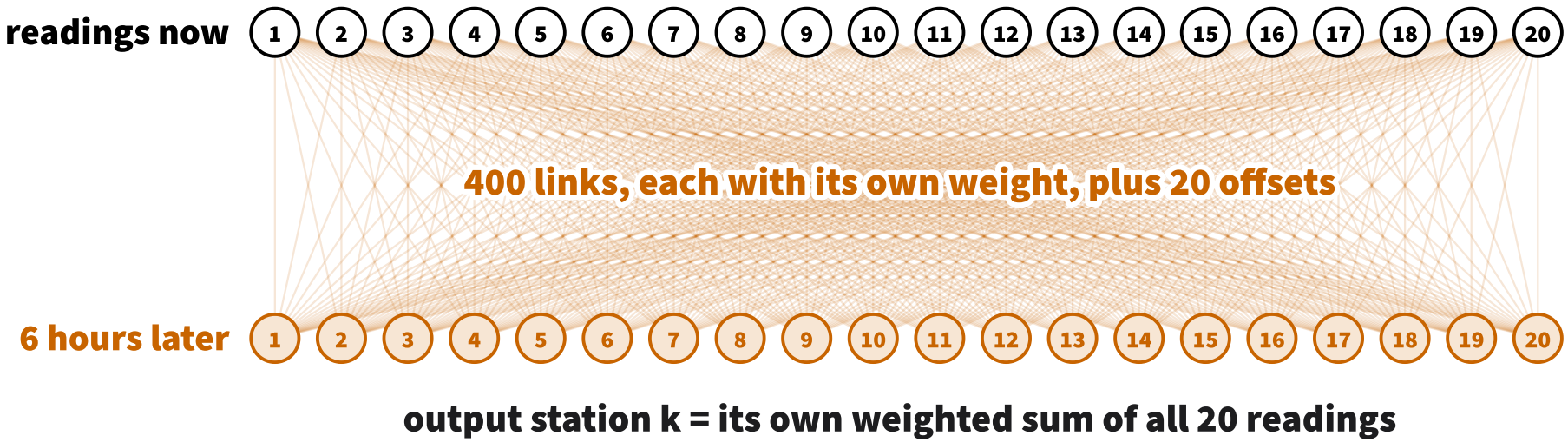
the patch drifts east and spreads out

SO **The rule is local: a station needs only its two neighbours. The forecaster is never told it. It sees only 100 noisy pairs of readings, 6 hours apart, and in every pair of this season the warm patch sits over stations 4 to 8.**

NEXT Try the obvious network first. What does it learn?

2 · The trouble: a plain network reads all 20 stations

Try the obvious network first. It is not told about the ring: in the picture every reading feeds every forecast, so each forecast learns one weight for each of the 20 readings plus a constant, 20 × 21 = 420 numbers, fitted to the 100 pairs.



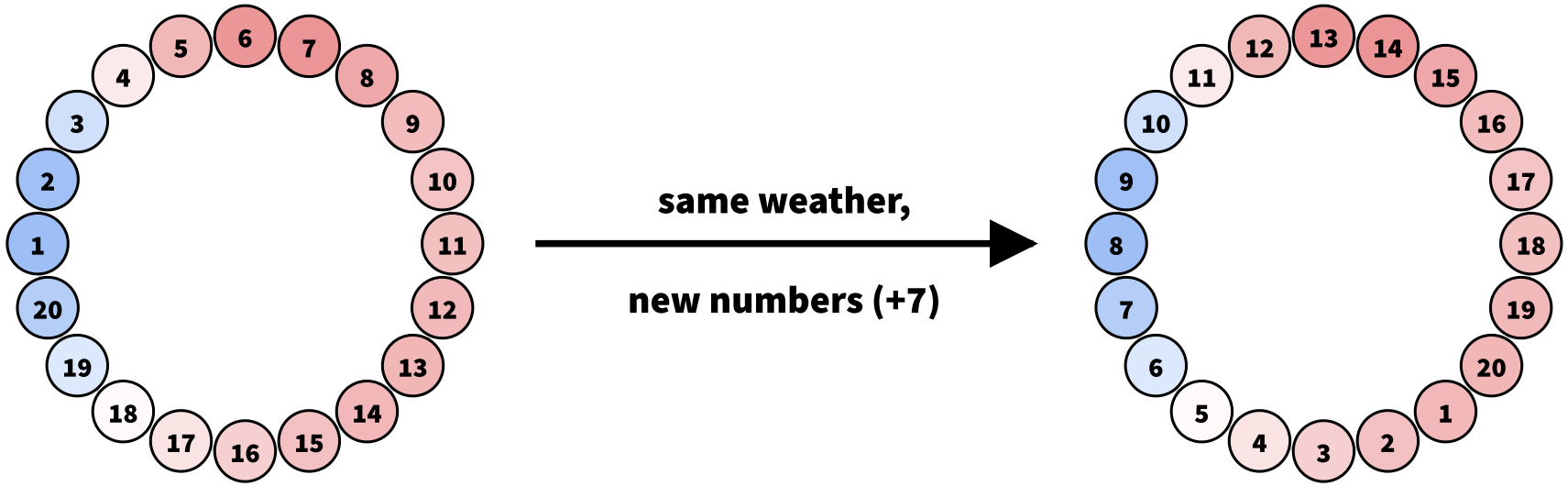
output station k = its own weighted sum of all 20 readings

SO **On new days of the same season its error is 0.05 °C: it has found something that fits the season, without ever learning which stations are neighbours.**

NEXT What if the same weather is numbered from another station?

2 · The trouble: renumber the stations

Same weather, but start counting from a different station: add 7 to every number. The patch that the training season called stations 4 to 8 is now called 11 to 15. Only the labels changed; the air is where it was.



numbers of the training season — renumbered

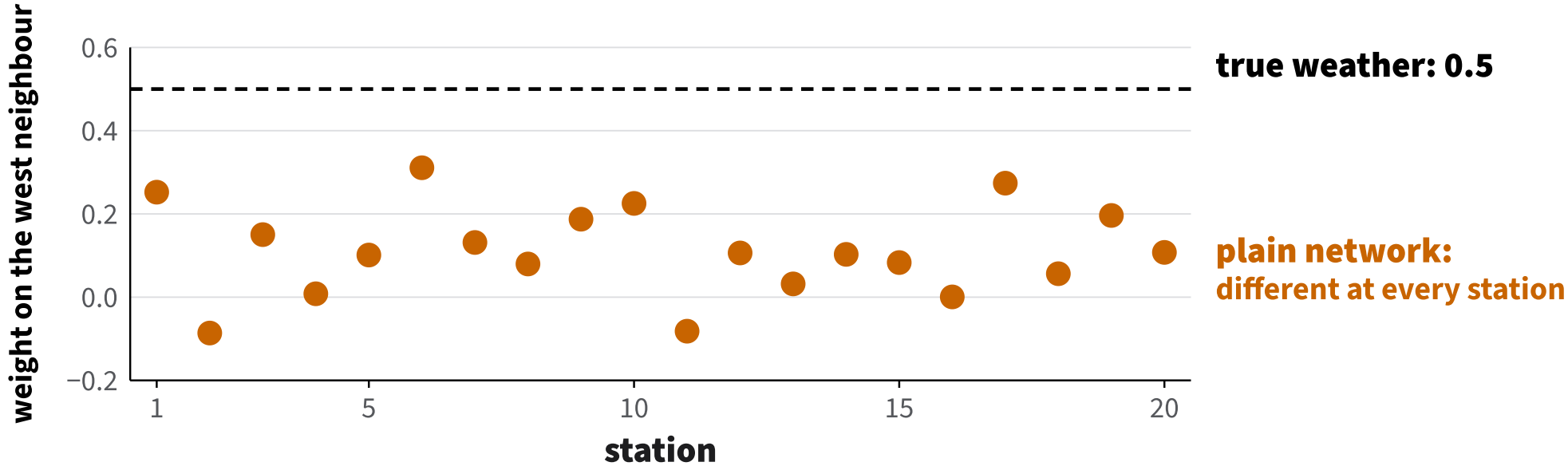
Q: Does the plain network still forecast well? A: No. Its error jumps from 0.05 °C to 1.37 °C.

SO **Worse than keeping today's values (0.25 °C): it failed.**

NEXT What did it learn instead?

2 · Why: each station number got its own rule

Ask each of the 20 forecasts the same question: if your west neighbour were 1 °C warmer, how much would you rise? The true weather answers 0.5 at every station.

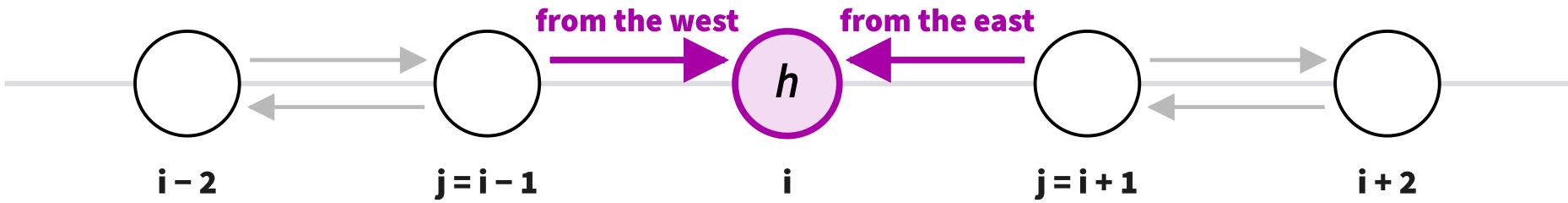


SO **Every station gives a different answer and none of them is 0.5. Each forecast has its own weights, so the network learned where the patch sat this season instead of the rule. It also has exactly 20 input slots, so 100 stations do not fit.**

NEXT Can we force one rule on every station?

3 · The fix: one shared message rule on every link

Force every station to use the same rule. A station adds one message from each neighbour to its own temperature.



grey: the same rule M_θ sends messages on every other link too

$$h_i \leftarrow h_i + \sum_{j \in N(i)} M_\theta(h_i, h_j, e_{ji})$$

h_i: temperature of station i · **N(i)**: its two neighbours · **e_{ji}**: west or east · **M_θ**: a small network, 81 numbers θ

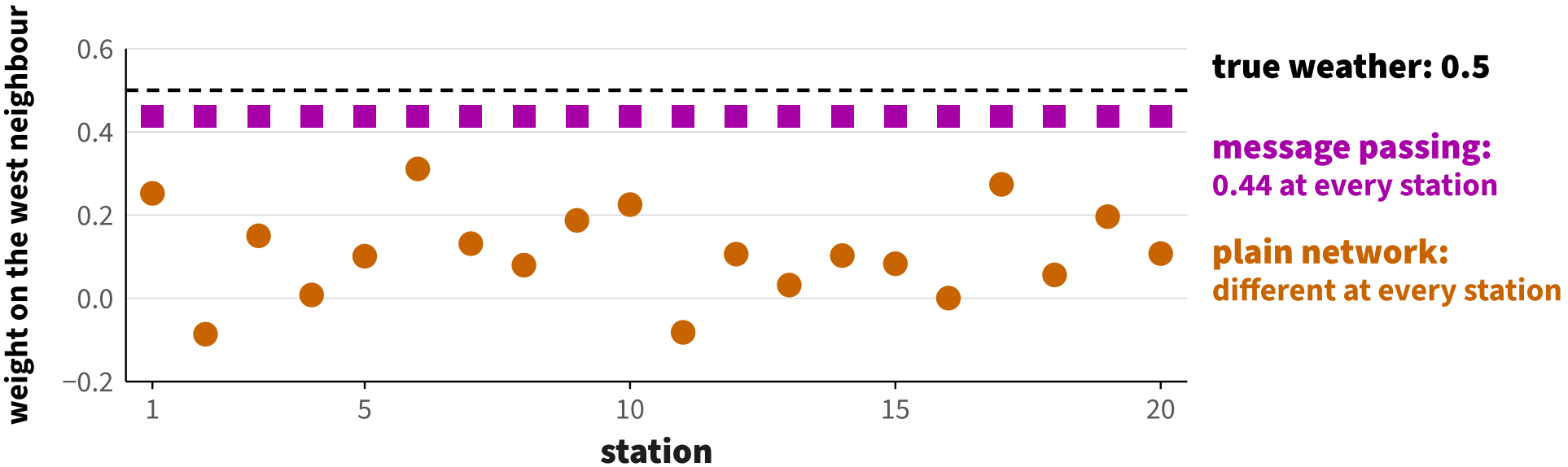
M_θ sees two temperatures and which side the neighbour is on. It never sees a station number.

SO **One round of messages is one 6-hour step.**

NEXT Trained on the same 100 pairs, which rule does it learn?

3 · The fix: every station now uses the same rule

Train M_θ on the same 100 pairs. Then ask every station the question from the last slide: if the west neighbour were 1 °C warmer, how much would you rise?

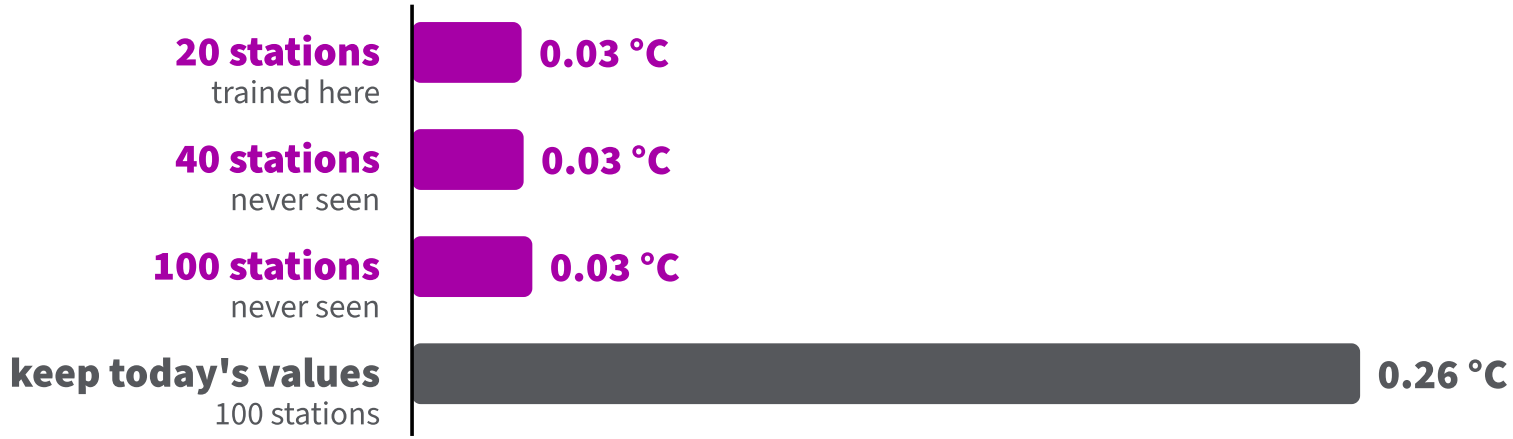


SO **Every station now answers 0.44, because every station uses the same M_θ. The true answer is 0.5. The readings carry noise, and noise in an input pulls its fitted weight toward zero: on the same 100 pairs without noise, the training gives 0.49.**

NEXT Can a rule trained on 20 stations forecast 100?

4 · The result: trained on 20 stations, run on 100

Keep those 81 numbers and run them on a ring of 100 stations. We can, because M_θ takes one neighbour at a time, and on any ring a station has two neighbours. Nothing in the rule counts the stations.



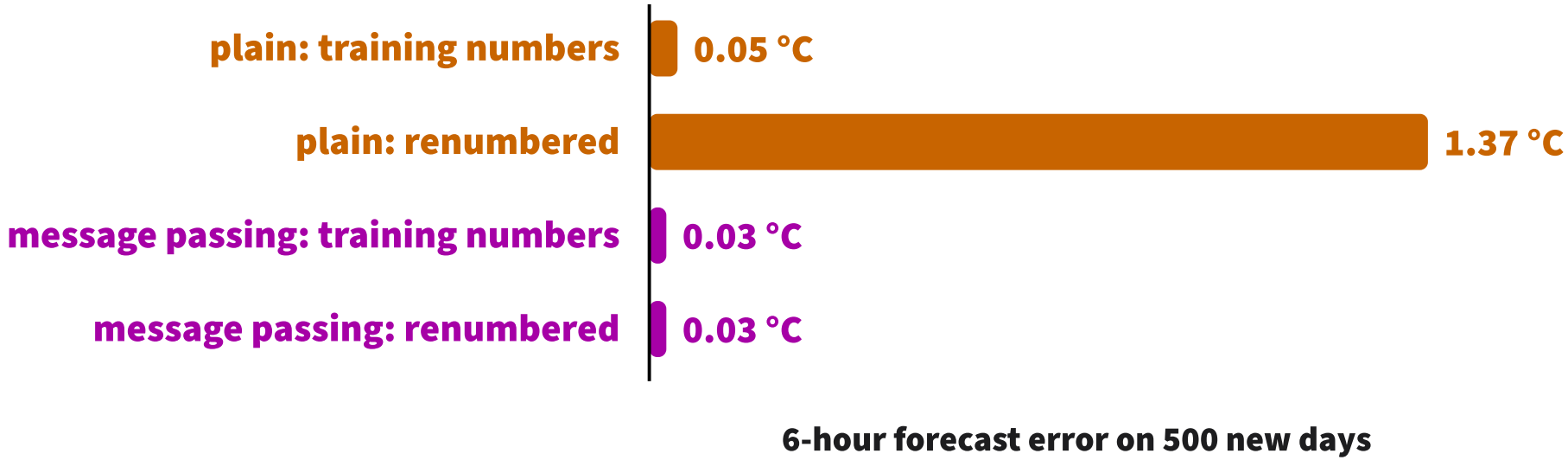
Q: Is the forecast on 100 stations worse than on 20? A: Hardly: 0.03 °C on both.

SO **A rule about neighbours ignores the number of stations. The plain network cannot even be tried here: its 420 weights need exactly 20 readings.**

NEXT And when the stations are renumbered?

4 · The result: renumbering cannot fool shared messages

Renumber by 7 again. The rule uses three things: a station's own temperature, its two neighbours' temperatures, and which side each neighbour is on. Renumbering changes none of them.

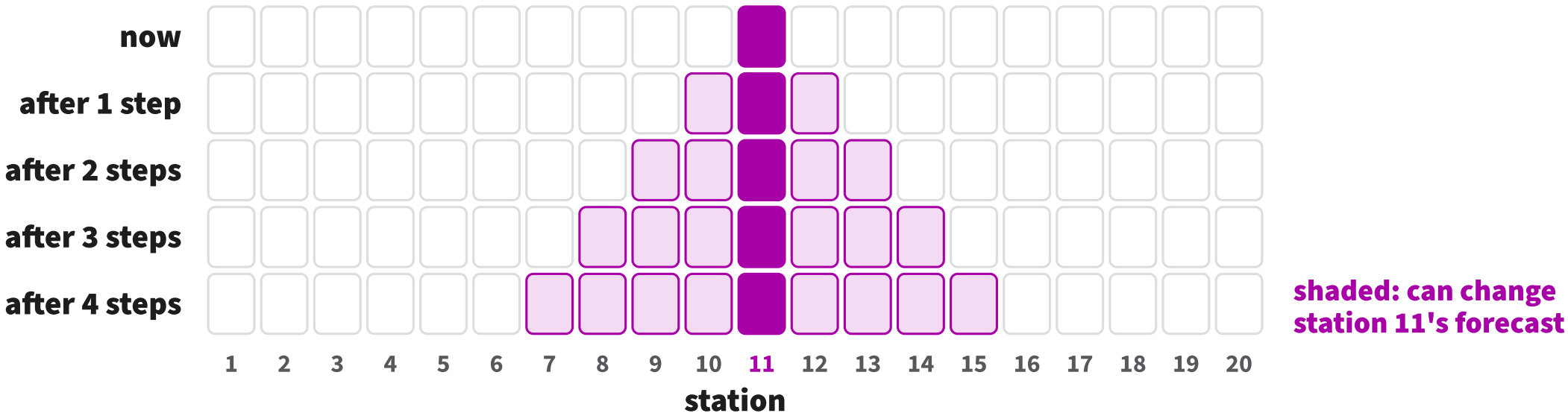


SO **Each station computes what it computed before, under a new name, so the forecast moves along with the stations: difference 0.000. The algebra is on the backup slide.**

NEXT How far can news travel in one round?

4 · The limit: messages travel one station per round

A message crosses one link per round. Which stations can change station 11’s forecast after 1, 2, 3 and 4 rounds?

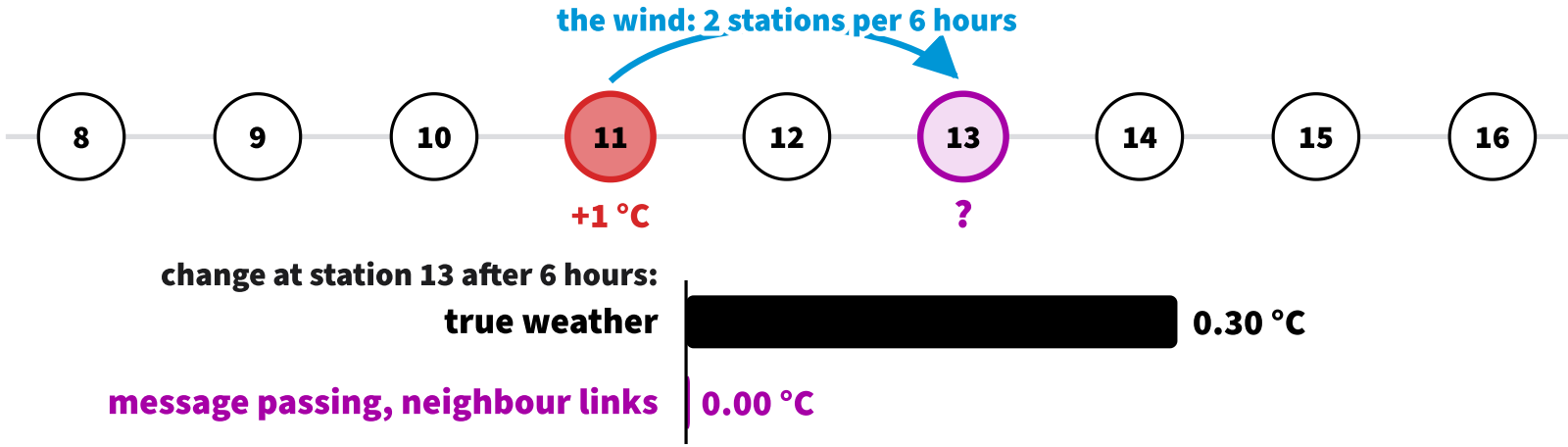


SO **After T rounds only the stations within T links can matter. That set is called the receptive field.**

NEXT What if the wind is faster than one station per step?

4 · The limit: a fast wind outruns the messages

Make the wind twice as fast: in 6 hours the air travels two stations east, so a station's new temperature comes mostly from the station two to its west: 0.3 · (two west) + 0.2 · (one west) + 0.3 · own + 0.2 · east. Warm station 11 by 1 °C.



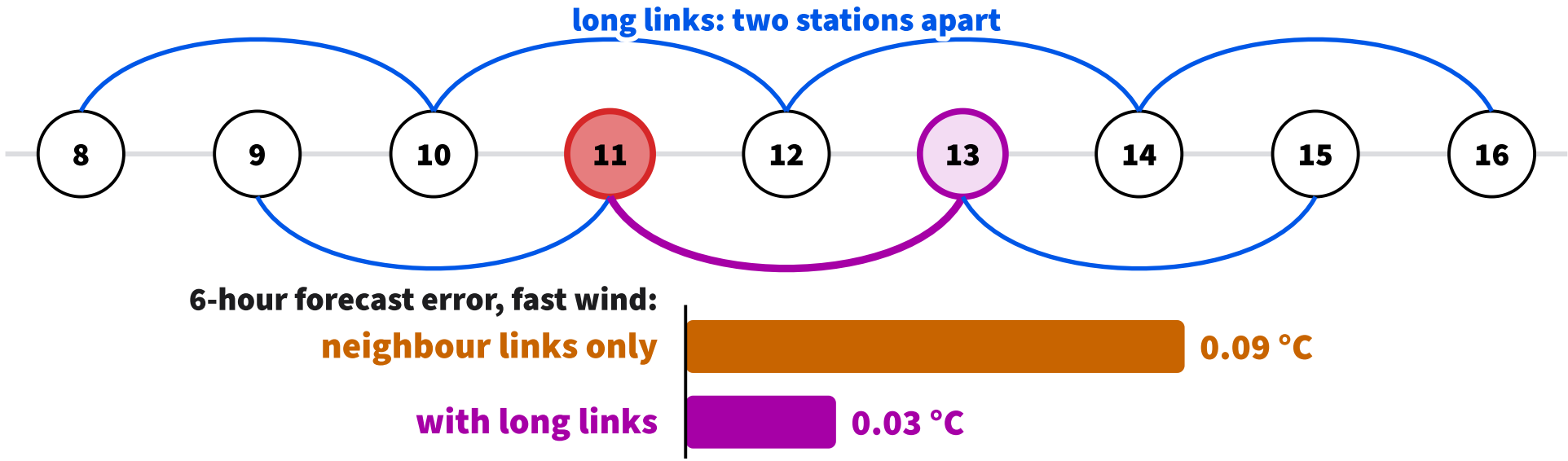
Q: Does station 13's forecast change in 6 hours? A: No. The true weather at station 13 rises by 0.30 °C, but one round of messages reaches only 12 and 14.

SO **Our forecaster passes one round per 6 hours, so news from two stations away is always one step behind the air.**

NEXT How can news travel two stations in one round?

5 · The fix for distance: add long links

Give each station two more links, to the stations two away. The news that needed two rounds now crosses one link.

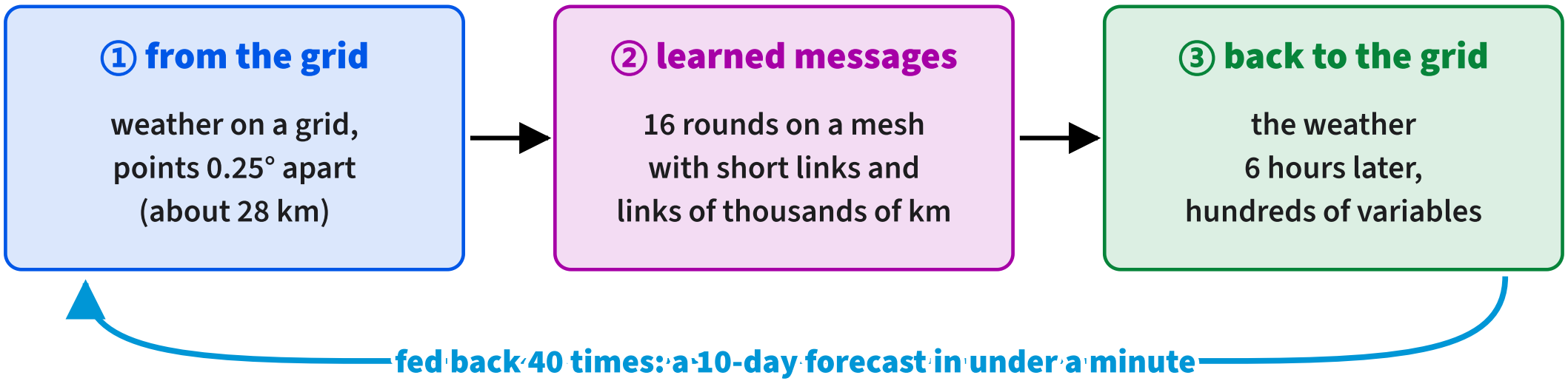


SO **Long links carry far news in one round: the error drops from 0.09 °C to 0.03 °C.**

NEXT What does the real system look like?

5 · GraphCast: the same recipe for the whole Earth

GraphCast (Google DeepMind, 2023) forecasts the globe in 6-hour steps of learned messages.



SO **Trained on 39 years of past weather, it beat the leading physics-based forecast on 90% of 1380 tests.**

NEXT Which ideas of Lecture 9 did we use?

5 · The same mathematics as Lecture 9

Every step of today's story uses an idea from the lectures.

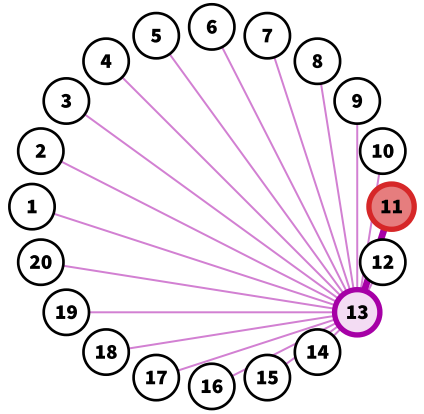
in today's lab	the lecture idea	where
stations and their neighbours	a graph: nodes, links, neighbours N(i)	Lecture 9
one message per link, summed, added	a message-passing layer: message, sum, update	Lectures 2, 9
the same M_θ on 20 or 100 stations	one trained network fits every graph size	Lecture 9
renumbering moves the forecast along	permutation equivariance	Lecture 9
T steps reach T stations	the receptive field: T layers see T links	Lecture 9
fitting θ to 100 noisy pairs	least squares: likelihood with Gaussian noise	Lecture 5
long links on a coarse ring	multiscale networks	Lecture 11

SO **Our forecaster is a graph neural network on a ring.**

NEXT What if a station could read every station at once?

5 · The frontier: let every station read every station

Replace the message rule by attention over all 20 stations, as in a Transformer: each station scores every other, softmax, weighted sum.



station 13 attends to all 20 stations;
the thick link: its weight on the warm station 11

neighbour links per round: 40

attention weights per round: 20 × 20 = 400

the fast wind (two stations per step)
arrives in one round: no long links needed

which links matter is decided per day,
by the attention weights (Lecture 9, Part 7)

Q: Does the fast wind's news now arrive in one round? A: Yes: every station is now one link from every other. The price: one weight for every pair, 20 × 20 = 400, instead of 20 × 2 = 40 links.

SO **Attention is message passing on the complete graph (Lecture 9, Part 7). GraphCast chose long links; a Transformer chooses all links. Same network, different graph.**

NEXT The argument in three sentences.

Conclusion

TROUBLE A plain network learned one rule per station number: renumbering broke it, and 100 stations cannot enter.

FIX One shared message rule on every link works on **any size and any numbering**.

LIMIT After T rounds a station has heard only the stations within T links, so far news needs **long links**.

FRONTIER Link every station to every station and the message rule becomes **attention**: a Transformer is this network on the complete graph.

Remember: build the symmetry of the problem into the network, then check how far and how long it can see.

Your turn

Write your guess first, then run it: `python lab9_explore.py` (Canvas Files → Lab 9).

Each experiment is one command and prints one number; the three together are the argument of this lab.

1. **Any size, any numbering.** Train on 20 stations; run on 100, and on a renumbered ring. Does the error grow?
2. **How far it sees.** Warm one station and count the rounds until a station two links away moves. Does the fast wind outrun the messages?
3. **The fix.** Add the long links. How much of the error do they remove?

Then push it: the kit also runs the four experiments in the backup slides, from forty rounds of averaging to ten days of data. Bring one surprising number to Thursday's lecture.

Selected References (1 of 3): the weather story

Battaglia, P. W., Hamrick, J. B., Bapst, V., et al. (2018).
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arXiv:1806.01261.

Gilmer, J., Schoenholz, S. S., Riley, P. F., Vinyals, O., and Dahl, G. E. (2017).
Neural message passing for quantum chemistry.
International Conference on Machine Learning (ICML), Proceedings of Machine Learning Research 70:1263–1272.

Hersbach, H., Bell, B., Berrisford, P., et al. (2020).
The ERA5 global reanalysis.
Quarterly Journal of the Royal Meteorological Society, 146(730):1999–2049.

Kochkov, D., Yuval, J., Langmore, I., et al. (2024).
Neural general circulation models for weather and climate.
Nature, 632:1060–1066.

Lam, R., Sanchez-Gonzalez, A., Willson, M., et al. (2023).
Learning skillful medium-range global weather forecasting.
Science, 382(6677):1416–1421.

Price, I., Sanchez-Gonzalez, A., Alet, F., et al. (2025).
Probabilistic weather forecasting with machine learning.
Nature, 637:84–90.

Selected References (2 of 3): the frontier story

Di Giovanni, F., Giusti, L., Barbero, F., Luise, G., Liò, P., and Bronstein, M. M. (2023).
On over-squashing in message passing neural networks: the impact of width, depth, and topology.
International Conference on Machine Learning (ICML), PMLR 202:7865–7885.

Dwivedi, V. P., Luu, A. T., Laurent, T., Bengio, Y., and Bresson, X. (2022).
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International Conference on Learning Representations (ICLR).

Garcia Satorras, V. and Welling, M. (2021).
Neural enhanced belief propagation on factor graphs.
International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 130:685–693.

Garg, V. K., Jegelka, S., and Jaakkola, T. (2020).
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International Conference on Machine Learning (ICML), PMLR 119:3419–3430.

Selected References (3 of 3): the frontier story

Oono, K. and Suzuki, T. (2020).
Graph neural networks exponentially lose expressive power for node classification.
International Conference on Learning Representations (ICLR).

Rampášek, L., Galkin, M., Dwivedi, V. P., Luu, A. T., Wolf, G., and Beaini, D. (2022).
Recipe for a general, powerful, scalable graph Transformer.
Advances in Neural Information Processing Systems 35.

Topping, J., Di Giovanni, F., Chamberlain, B. P., Dong, X., and Bronstein, M. M. (2022).
Understanding over-squashing and bottlenecks on graphs via curvature.
International Conference on Learning Representations (ICLR).

Xu, K., Hu, W., Leskovec, J., and Jegelka, S. (2019).
How powerful are graph neural networks?
International Conference on Learning Representations (ICLR).

Thank you!

Next: Thursday's lecture, 12 November

Backup: the toy weather in symbols, training M_θ, why renumbering cannot change the forecast, and four more ways to push the ring.

Backup: the toy weather in symbols

Spreading plus wind, one step of 6 hours.

u_i(t): temperature of station i (°C above normal) after t steps · D: spreading strength · c: wind strength (stations per step) · stations i − 1, i + 1: the west and east neighbours.

$$u_i(t+1) = u_i + D \cdot (u_{i-1} - 2 \cdot u_i + u_{i+1}) - c \cdot (u_i - u_{i-1})$$

> **spreading evens out neighbours; the wind brings the west value in**

$$D = 0.2,\ c = 0.3: \quad u_i(t+1) = 0.5 \cdot u_{i-1} + 0.3 \cdot u_i + 0.2 \cdot u_{i+1}$$

> **collect the terms of each station**

$$0.5 + 0.3 + 0.2 = 1,\ \text{all} \geq 0$$

> **a weighted average: never above the warmest station, never below the coldest (stable); the ring's average is kept**

Fast wind: replace u_{i-1} in the wind term by u_{i-2}: weights 0.3, 0.2, 0.3, 0.2 on stations i − 2, i − 1, i, i + 1.

Backup: training the message rule

A small network, fitted by gradient steps on squared error.

$$M_\theta(h_i, h_j, e) = 8 \cdot \left(w^\top \tanh(A \cdot (h_i/8,\ h_j/8,\ e) + b) + b_0\right)$$

> A: 16 × 3, b and w: 16 each, b_0: 1, so θ has 81 numbers; e = +1 (west), −1 (east)

$$\mathcal{L}(\theta) = \tfrac{1}{100 \cdot 20} \sum_{k=1}^{100} \sum_{i=1}^{20} \left(\hat{u}_i^{(k)} - y_i^{(k)}\right)^2$$

> $\hat{u}_i^{(k)}$: the one-step forecast from reading k; $y_i^{(k)}$: the reading 6 hours later

$$\theta \leftarrow \theta - \eta \cdot (\text{Adam step on } \nabla_\theta \mathcal{L})$$

> 6000 steps, learning rate η = 0.01, gradients by the chain rule

Read off: at a calm state the learned weights are 0.44 (west), 0.36 (own), 0.20 (east); they sum to 0.99, not 1. The true weights are 0.5, 0.3, 0.2. Noise in the readings, not the number of pairs, causes the gap: without noise the same 100 pairs give 0.49, 0.30, 0.20, while with noise 2000 pairs still give 0.46, 0.34, 0.19. The west weight shrinks and the station's own value takes up the slack.

Backup: why renumbering cannot change the forecast

Renumbering moves stations and links together.

σ: the renumbering, station i becomes station σ(i) = i + 7 (counted around the ring) · $h'_{\sigma(i)} = h_i$: the renumbered readings.

$$j \in N(i) \iff \sigma(j) \in N(\sigma(i)), \quad e_{\sigma(j)\sigma(i)} = e_{ji}$$

> **a shift keeps every neighbour and every direction**

$$h'_{\sigma(i)} + \sum_{j' \in N(\sigma(i))} M_\theta(h'_{\sigma(i)}, h'_{j'}, e_{j'\sigma(i)}) = h_i + \sum_{j \in N(i)} M_\theta(h_i, h_j, e_{ji})$$

> **substitute j′ = σ(j): the same terms, the same sum**

So: new forecast at σ(i) = old forecast at i, exactly (Lecture 9, permutation equivariance). A mirror image would swap west and east, so it is not covered.

Backup · Push the ring: four more troubles

Five ways to push the ring, each breaking it in a way graph networks are known to break, and each with a fix. The last one was the frontier slide of the talk.

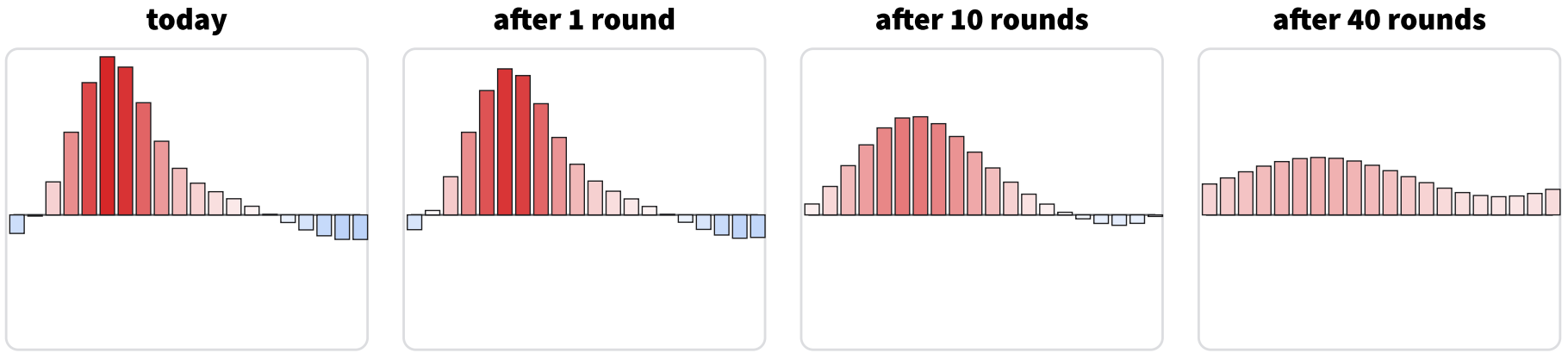
push it	what breaks	the fix
many more rounds	the weather fades out	keep each station's own reading
a single bridge	news cannot get through	add links across the bridge
one ring or two?	the forecaster cannot tell	give stations a position
only 10 days of data	learning falls behind a hand-written solver	keep the solver, learn the rest
every station reads every station	400 weights per round	attention: a Transformer

SO **We did the last one in the talk; here are the other four, each with a number you can reproduce.**

NEXT What if we pass messages 40 times?

Backup · Push 1: forty rounds, and the weather fades out

To see further, stack more rounds. Take the simplest round, each station ← the mean of itself and its two neighbours, and run it 40 times on one day.



each round: station ← mean of itself and its two neighbours; the spread shrinks by the same factor 0.9674 every round

Q: After 40 rounds, where is the warm patch? A: Fading: the spread fell from 5.8 °C to 1.3 °C, by the same factor 0.967 every round.

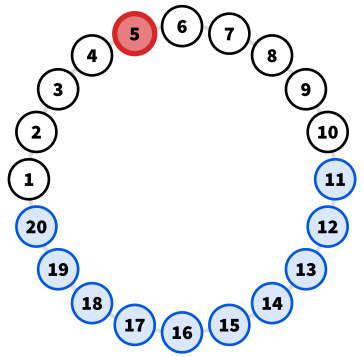
SO **Averaging alone erases the weather (over-smoothing). Our rule survives because each round keeps the station's own reading: h ← h + messages.**

NEXT What if the ring has a bottleneck?

Backup · Push 2: one bridge, and news cannot get through

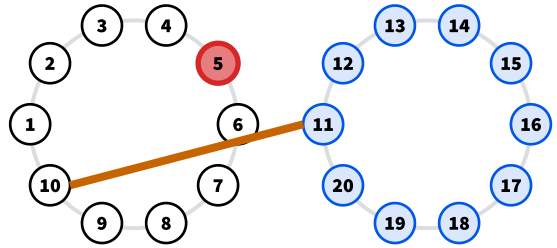
Cut the ring into two rings of 10 joined by one bridge: two valleys and one mountain pass. Warm station 5 by 1 °C.

one ring of 20



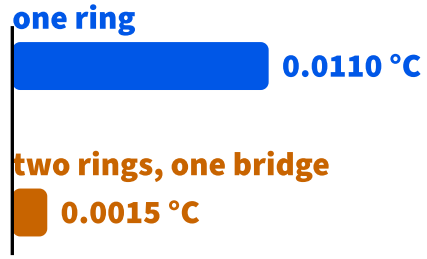
stations 11–20 (blue): two ways in

two rings of 10, one bridge



stations 11–20: one way in, the bridge 10–11

news from the +1 °C bump at station 5 reaching stations 11–20 after 6 rounds



7 times less across the bridge

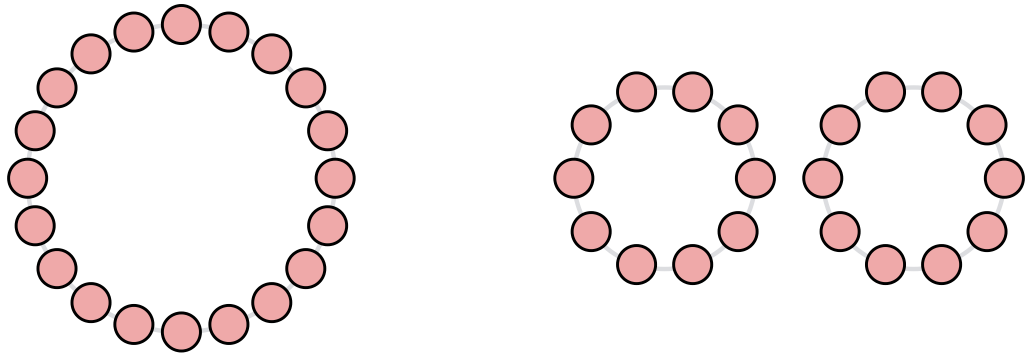
Q: After 6 rounds, how much of that news reaches the other valley? A: 0.0015 °C across the bridge, against 0.011 °C on the full ring: 7 times less.

so **All the news of one valley must squeeze through one link (over-squashing). The fix: add links across the pass, as we added long links in Part 4.**

NEXT Can the forecaster tell one ring from two?

Backup · Push 3: one ring or two? the forecaster cannot tell

Give the trained rule a uniform day, +2 °C everywhere, once on the ring of 20 and once on two separate rings of 10.



one ring of 20, every station +2.0 °C two rings of 10, every station +2.0 °C

forecast at every station

one ring 2.00 °C

two rings 2.00 °C

difference 0.000: identical

every station sees two neighbours at the same temperature: same messages, same forecast

the true weather also stays at 2.0 °C on both graphs: for local physics this limit costs nothing

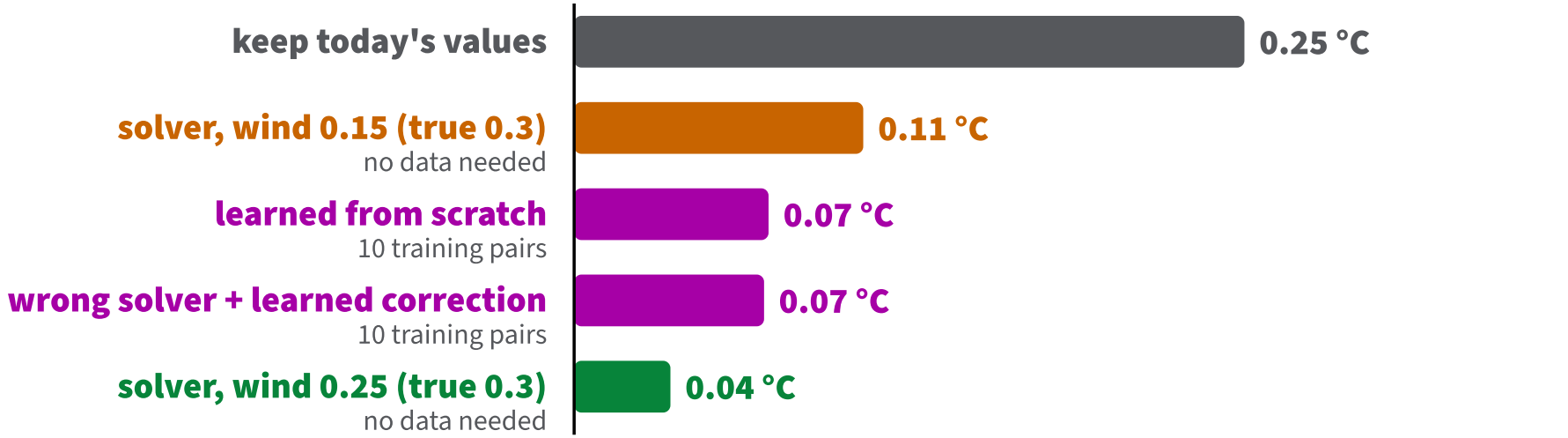
Q: Do the two forecasts differ? A: No: identical at every station, difference 0.000.

SO **Every station sees two neighbours at the same temperature, so it gets the same messages: the rule cannot count stations or see the shape of the graph. For local weather this costs nothing; for a global question, give each station a position input.**

NEXT What if we only have ten days of data?

Backup · Push 4: ten days of data, against a hand-written solver

Now only 10 training pairs, not 100. The alternative is a solver written from the physics, with the wind speed wrong: badly (0.15) or nearly right (0.25); the truth is 0.3.



6-hour forecast error on 500 new days; the lab's rule, trained on 100 pairs, reaches 0.03 °C

Q: Which forecasts best: learn from the 10 days, or keep a solver? A: The nearly-right solver, 0.037 °C; learning gives 0.074 °C; the badly wrong solver 0.110 °C.

SO **With little data, keep a known rule that is nearly right and learn only when it is badly wrong; with 100 days, learning wins (0.030 °C). NeuralGCM keeps the physics solver and learns the rest.**

NEXT The four pushes on one page, with their names.

Backup · Four pushes, four names, four things to try

push it	its name (Lecture 9)	on our ring	try it, extending lab9_explore.py
40 rounds	over-smoothing (p. 65)	spread × 0.967 per round	add h ← h + messages; does the patch survive?
one bridge	over-squashing (p. 67)	7× less news	add a second bridge; measure the news again
one ring or two	colour-refinement limit (p. 74)	difference 0.000	add a relative position input; can it count stations?
10 days of data	generalisation (p. 46)	0.074 vs 0.030 °C	vary the pairs from 5 to 100; when does learning beat the 0.25 solver?

SO **Every frontier result is a push our ring can show, with a name and a fix; the papers are on the reference pages.**

NEXT The kit reproduces every number above.